# Use Cases

for

# PathSG

**Version 1.2**

**Prepared by Max Lian**

**PathSG**

**26 Oct 2026**

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# Revision History

| Name | Date | Reason For Changes | Version |
|---|---|---|---|
| Max Lian | 05 Sep 2026 | Initial Use Case Description | V1.0 |
| Max Lian | 12 Sep 2026 | Additional Use Case Description | V1.1 |
| Max Lian | 26 Oct 2026 | Revised “3.1: Generate Education Roadmap”<br>Revised “5.1: Update School Database”<br>Added “5.2: Export School Database” | V1.2 |

# Guidance for Use Case Template

Document each use case using the template shown in the Appendix. This section provides a description of each section in the use case template.

# 1. Use Case Identification

## 1.1. Use Case ID

Give each use case a unique numeric identifier, in hierarchical form: X.Y. Related use cases can be grouped in the hierarchy. Functional requirements can be traced back to a labeled use case.

## 1.2. Use Case Name

State a concise, results-oriented name for the use case. These reflect the tasks the user needs to be able to accomplish using the system. Include an action verb and a noun. Some examples:

- View part number information.
- Manually mark hypertext source and establish link to target.
- Place an order for a CD with the updated software version.

## 1.3. Use Case History

### 1.3.1 Created By

Supply the name of the person who initially documented this use case.

### 1.3.2 Date Created

Enter the date on which the use case was initially documented.

### 1.3.3 Last Updated By

Supply the name of the person who performed the most recent update to the use case description.

### 1.3.4 Date Last Updated

Enter the date on which the use case was most recently updated.

# 2. Use Case Definition

## 2.1. Actor

An actor is a person or other entity external to the software system being specified who interacts with the system and performs use cases to accomplish tasks. Different actors often correspond to different user classes, or roles, identified from the customer community that will use the product. Name the actor(s) that will be performing this use case.

## 2.2. Description

Provide a brief description of the reason for and outcome of this use case, or a high-level description of the sequence of actions and the outcome of executing the use case.

## 2.3. Preconditions

List any activities that must take place, or any conditions that must be true, before the use case can be started. Number each precondition. Examples:

1. User's identity has been authenticated.
2. User's computer has sufficient free memory available to launch task.

## 2.4. Postconditions

Describe the state of the system at the conclusion of the use case execution. Number each postcondition. Examples:

1. Document contains only valid SGML tags.
2. Price of item in database has been updated with new value.

## 2.5. Priority

Indicate the relative priority of implementing the functionality required to allow this use case to be executed. The priority scheme used must be the same as that used in the software requirements specification.

## 2.6. Frequency of Use

Estimate the number of times this use case will be performed by the actors per some appropriate unit of time.

## 2.7. Flow of Events

Provide a detailed description of the user actions and system responses that will take place during execution of the use case under normal, expected conditions. This dialog sequence will ultimately lead to accomplishing the goal stated in the use case name and description. This description may be written as an answer to the hypothetical question, "How do I <accomplish the task stated in the use case name>?" This is best done as a numbered list of actions performed by the actor, alternating with responses provided by the system.

## 2.8. Alternative Flows

Document other, legitimate usage scenarios that can take place within this use case separately in this section. State the alternative course, and describe any differences in the sequence of steps that take place. Number each alternative course using the Use Case ID as a prefix, followed by "AC" to indicate "Alternative Course". Example: X.Y.AC.1.

## 2.9. Exceptions

Describe any anticipated error conditions that could occur during execution of the use case, and define how the system is to respond to those conditions. Also, describe how the system is to respond if the use

case execution fails for some unanticipated reason. Number each exception using the Use Case ID as a prefix, followed by "EX" to indicate "Exception". Example: X.Y.EX.1.

## 2.10. Includes

List any other use cases that are included ("called") by this use case. Common functionality that appears in multiple use cases can be split out into a separate use case that is included by the ones that need that common functionality.

## 2.11. Special Requirements

Identify any additional requirements, such as nonfunctional requirements, for the use case that may need to be addressed during design or implementation. These may include performance requirements or other quality attributes.

## 2.12. Assumptions

List any assumptions that were made in the analysis that led to accepting this use case into the product description and writing the use case description.

## 2.13. Notes and Issues

List any additional comments about this use case or any remaining open issues or TBDs (To Be Determineds) that must be resolved. Identify who will resolve each issue, the due date, and what the resolution ultimately is.

# Use Case Template

| | | | |
|---|---|---|---|
| Use Case ID: | | | |
| Use Case Name: | | | |
| Created By: | | Last Updated By: | |
| Date Created: | | Date Last Updated: | |

| | |
|---|---|
| Actor: | |
| Description: | |
| Preconditions: | |
| Postconditions: | |
| Priority: | |
| Frequency of Use: | |
| Flow of Events: | |
| Alternative Flows: | |
| Exceptions: | |
| Includes: | |
| Special Requirements: | |
| Assumptions: | |
| Notes and Issues: | |

| Use Case ID: | UC-1.1 | | |
|---|---|---|---|
| Use Case Name: | Search and Filter School Directory | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 04 Sep 2026 | Date Last Updated: | 04 Sep 2026 |

| Actor: | All Users (Initiating Actor) |
|---|---|
| Description: | Actors are able to search the school directory and apply relevant filters (e.g., governance, education sector, courses offered) to narrow the search results. |
| Preconditions: | - The actor is on the school directory page. |
| Postconditions: | - A filtered and sorted search result or an empty result if there are no matches. |
| Priority: | Very High |
| Frequency of Use: | Very High |
| Flow of Events: | 1. The user navigates to the school directory page.<br>2. The system displays a complete view of the schools.<br>3. The user optionally enters a keyword or sets filters to narrow down the search results.<br>4. The system sanitizes user input and queries the database.<br>5. The system displays a filtered view of the schools.<br>6. The user optionally changes the sort order.<br>7. The system displays a sorted view of the results. |
| Alternative Flows: | - AF-S4. If the keyword contains special characters, the system will treat it as a literal and query for its exact match. Return to S5.<br>- AF-S5. If there are no search results, the system will display "No schools match your search criteria. Try clearing filters or adjusting your search keywords." with a “Reset All Filters” button. Return to S3. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error and return an empty table. |
| Includes: | None |
| Special Requirements: | - PERF-4: The system shall load and render the school directory within 2 seconds on visit or query.<br>- QUAL-1: The school directory view shall be responsive and adjust to the screen size of the user.<br>- SEC-4: The system shall sanitize the user input on the backend. |
| Assumptions: | 1. The school database is already populated. |
| Notes and Issues: | None |

| Use Case ID: | UC-1.2 | | |
|---|---|---|---|
| Use Case Name: | View School Profile | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 04 Sep 2026 | Date Last Updated: | 04 Sep 2026 |

| Actor: | All Users (Initiating Actor) |
|---|---|
| Description: | Actors are able to view the school profile that contains additional information that is not visible from UC-1.1's search view. |
| Preconditions: | - The actor selects a school from the school directory page. |
| Postconditions: | - A populated profile page is displayed. |
| Priority: | High |
| Frequency of Use: | High |
| Flow of Events: | 1. The user selects a school from UC-1.1.<br>2. The system queries for the full school record.<br>3. The system displays the following information:<br>o School Identity (e.g., name, logo, governance)<br>o Location (e.g., address, postal code)<br>o Course & Curriculum<br>o Itemized School Fees<br>o Financial Aid Programs (e.g., tuition grant, bursary)<br>o Scholarship Details (e.g., eligibility requirements)<br>o Historical Statistics (e.g., admission cut-off scores)<br>o User Reviews |
| Alternative Flows: | - AF-S3. If a section has no data, the system displays "not available". Resume with S3. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | - UC-4.2 |
| Special Requirements: | - BR-1: The system shall display course fee breakdowns based on citizenship status.<br>- PERF-1: The system shall load and render the school profile within 2 seconds on visit or query.<br>- QUAL-1: The school profile view shall be responsive and adjust to the screen size of the user. |
| Assumptions: | 1. The school database is already populated. |
| Notes and Issues: | None |

| Use Case ID: | UC-2.1 | | |
|---|---|---|---|
| Use Case Name: | Register User Account | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 04 Sep 2026 | Date Last Updated: | 04 Sep 2026 |

| Actor: | Guest Users (Initiating Actor) |
|---|---|
| Description: | Actors are able to create a user account to save their preferences. |
| Preconditions: | - The actor is not authenticated.<br>- The actor is on the registration page. |
| Postconditions: | - A user account is created and the actor is logged in; or<br>- No user account is created and the form shows an inline error. |
| Priority: | High |
| Frequency of Use: | Low |
| Flow of Events: | 1. The guest user navigates to the registration page.<br>2. The system displays the registration form containing username and password fields.<br>3. The guest user enters a username and password.<br>4. The guest user clicks on the “Register” button.<br>5. The system sanitizes user input.<br>6. The system checks that the username is not already used.<br>7. The system checks that the password is sufficiently complex.<br>8. The system hashes the password and creates the account.<br>9. The system creates a login session and notifies the guest user. |
| Alternative Flows: | - AF-S5: If the user input contains special characters, an inline error "Username cannot contain special characters. Please use only letters and numbers." will be displayed on the form. Return to S3.<br>- AF-S6: If the username is already used, an inline error "Username is already taken. Please choose a different username." will be displayed on the form. Return to S3.<br>- AF-S7: If the password is not sufficiently complex, an inline error "Password must be at least 8 characters long and contain at least one number and one special character." will be displayed on the form. Return to S3. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | - PERF-5: The system shall process the authentication request and return a response within 1s.<br>- QUAL-2: The system shall show the inline errors within 0.3s.<br>- SEC-1: The system shall not request, collect, or store any personally identifiable information (PII) in the database.<br>- SEC-2: The system shall enforce password complexity (min. of 8 characters, 1 numeral, 1 special character) on the backend.<br>- SEC-3: The system shall hash the password on the backend.<br>- SEC-4: The system shall sanitize the user input on the backend.<br>- SEC-5: The system shall handle user sessions using JSON Web Tokens (JWT) stored in HTTP-only cookies. |

| | |
|---|---|
| Assumptions: | 1. The username is the unique identifier of a user account. |
| Notes and Issues: | Considerations<br>1. An optional email field for password recovery.<br>2. Additional password security measures such as salting. |

| Use Case ID: | UC-2.2 | | |
|---|---|---|---|
| Use Case Name: | Login User Account | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 04 Sep 2026 | Date Last Updated: | 04 Sep 2026 |

| Actor: | Guest Users (Initiating Actor). |
|---|---|
| Description: | Actors are able to authenticate into their user account to access their saved preferences. |
| Preconditions: | - The actor is not authenticated.<br>- The actor has an existing user account.<br>- The actor is on the login page. |
| Postconditions: | - A user account is authenticated and the actor is logged in; or<br>- The actor is not logged in and the form shows an inline error. |
| Priority: | High |
| Frequency of Use: | Medium |
| Flow of Events: | 1. The guest user navigates to the login page.<br>2. The system displays the login form containing username and password fields.<br>3. The guest user enters a username and password.<br>4. The guest user clicks on the “Login” button.<br>5. The system sanitizes user input.<br>6. The system hashes the password and compares it with the stored hash for the username.<br>7. The system creates a login session and notifies the guest user. |
| Alternative Flows: | - AF-S6: If the username does not exist or the password hash does not match, an inline error will be displayed on the form. Return to S3. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | - PERF-5: The system shall process the authentication request and return a response within 1s.<br>- QUAL-2: The system shall show the inline errors within 0.3s.<br>- SEC-4: The system shall sanitize the user input on the backend.<br>- SEC-5: The system shall handle user sessions using JSON Web Tokens (JWT) stored in HTTP-only cookies. |
| Assumptions: | 1. The username is the unique identifier of a user account. |
| Notes and Issues: | Considerations<br>1. Additional account security measures such as rate limits. |

| Use Case ID: | UC-2.3 | | |
|---|---|---|---|
| Use Case Name: | Logout User Account | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 04 Sep 2026 | Date Last Updated: | 04 Sep 2026 |

| Actor: | Authenticated Users (Initiating Actor). |
|---|---|
| Description: | Actors are able to logout from their user account and return to a guest user. |
| Preconditions: | - The actor is authenticated. |
| Postconditions: | - The session is terminated and the actor returns to guest user. |
| Priority: | High |
| Frequency of Use: | Low |
| Flow of Events: | 1. The authenticated user clicks on the “Logout” button.<br>2. The system invalidates the session.<br>3. The system clears the JWT cookie.<br>4. The system redirects the guest user to root. |
| Alternative Flows: | - AF-S2: The session cannot be invalidated. The system proceeds to clear the JWT cookie. Resume with S3. |
| Exceptions: | None |
| Includes: | None |
| Special Requirements: | None |
| Assumptions: | 1. Removing the JWT cookie is sufficient to end the session. |
| Notes and Issues: | None |

| Use Case ID: | UC-2.4 | | |
|---|---|---|---|
| Use Case Name: | View User Profile | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 05 Sep 2026 | Date Last Updated: | 05 Sep 2026 |

| Actor: | Authenticated Users (Initiating Actor). |
|---|---|
| Description: | Actors are able to configure their profile to personalize their experience on the system. |
| Preconditions: | - The actor is authenticated.<br>- The actor is on the profile page. |
| Postconditions: | - The profile is saved; or<br>- The profile is not saved and the form shows an inline error. |
| Priority: | High |
| Frequency of Use: | Low |
| Flow of Events: | 1. The authenticated user navigates to the profile page.<br>2. The system displays the target education level selector.<br>3. The authenticated user selects a target education level.<br>4. The system displays the relevant fields for the education level.<br>5. The authenticated user enters their residential postal code and fills in the relevant fields for the education level.<br>6. The authenticated user clicks on the “Save” button.<br>7. The system sanitizes user input.<br>8. The system validates the postal code as a 6-digit string.<br>9. The system validates the academic score against the range.<br>10. The system saves the profile and notifies the authenticated user. |
| Alternative Flows: | - AF-S8: If the postal code is not valid, an inline error “Please enter a valid 6-digit Singapore postal code.” will be displayed on the form. Return to S5.<br>- AF-S9: If the academic score is not valid, an inline error “Invalid academic score!” will be displayed on the form. Return to S5. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | - BR-3: The system shall validate academic score boundaries (e.g., PSLE AL 4-32, O-Level L1R5 2-54).<br>- BR-4: The system shall validate postal code user input as a valid 6-digit Singapore postal code.<br>- PERF-2: The system shall render the education fields within 1s of choosing an education level.<br>- QUAL-2: The system shall show the inline errors within 0.3s.<br>- SEC-1: The system shall not request, collect, or store any personally identifiable information (PII) in the database.<br>- SEC-4: The system shall sanitize the user input on the backend. |
| Assumptions: | 1. The authenticated user is truthful with their saved preferences. |
| Notes and Issues: | None |

| Use Case ID: | UC-2.5 | | |
|---|---|---|---|
| Use Case Name: | Bookmark School | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 05 Sep 2026 | Date Last Updated: | 12 Sep 2026 |

| Actor: | Authenticated Users (Initiating Actor). |
|---|---|
| Description: | Actors are able to add a school bookmark so it persists in their saved list across sessions. |
| Preconditions: | - The actor is authenticated.<br>- The actor is on a school summary card or school profile page. |
| Postconditions: | - The school is added into the saved list; or<br>- The school is not added and the system shows a popup error. |
| Priority: | Medium |
| Frequency of Use: | Medium |
| Flow of Events: | 1. The authenticated user clicks on the bookmark icon.<br>2. The system adds the school into the saved list.<br>3. The system changes the bookmark icon to show new state. |
| Alternative Flows: | None |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | - BR-2: The system shall allow bookmark management function only to authenticated users. |
| Assumptions: | 1. There is no limit to the number of bookmarks.<br>2. The bookmark icon is not shown to guest users. |
| Notes and Issues: | None |

| Use Case ID: | UC-2.6 | | |
|---|---|---|---|
| Use Case Name: | Unbookmark School | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 05 Sep 2026 | Date Last Updated: | 12 Sep 2026 |

| Actor: | Authenticated Users (Initiating Actor). |
|---|---|
| Description: | Actors are able to remove a school bookmark so it no longer persists in their saved list across sessions. |
| Preconditions: | - The actor is authenticated.<br>- The school is bookmarked. |
| Postconditions: | - The school is removed from the saved list; or<br>- The school is not removed and the system shows a popup error. |
| Priority: | Medium |
| Frequency of Use: | Medium |
| Flow of Events: | 1. The authenticated user clicks on the bookmark icon.<br>2. The system removes the school from the saved list.<br>3. The system changes the bookmark icon to show new state. |
| Alternative Flows: | None |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | - BR-2: The system shall allow bookmark management function only to authenticated users. |
| Assumptions: | 1. There is no confirmation dialog for removing a bookmark.<br>2. There is no undo option after removing a bookmark. |
| Notes and Issues: | None |

| Use Case ID: | UC-2.7 | | |
|---|---|---|---|
| Use Case Name: | View Bookmarked Schools | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 05 Sep 2026 | Date Last Updated: | 12 Sep 2026 |

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|---|---|
| Actor: | Authenticated Users (Initiating Actor). |
| Description: | Actors are able to view all their bookmarked schools in one place. |
| Preconditions: | - The actor is authenticated.<br>- The actor is on the saved list page. |
| Postconditions: | - The saved list is displayed; or<br>- An empty state is displayed. |
| Priority: | Medium |
| Frequency of Use: | Medium |
| Flow of Events: | 1. The authenticated user navigates to the saved list page.<br>2. The system retrieves all the schools from the saved list.<br>3. The system displays a summary card for each school.<br>o School Name<br>o School Logo<br>o Governance<br>o Distance (computed as a straight line distance between the postal code of the authenticated user and the school)<br>o Entry Benchmark (admission cut-off scores) |
| Alternative Flows: | - AF-S2: If the saved list is empty, the system displays an empty state with a link back to the school directory.<br>- AF-S3: If there are no postal code saved, the system omits the distance. Resume with S3. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | - BR-2: The system shall allow bookmark management function only to authenticated users.<br>- PERF-3: The system shall compute the distance for all schools in the saved list within 2s.<br>- QUAL-1: The saved list view shall be responsive and adjust to the screen size of the user. |
| Assumptions: | 1. The saved list is of a realistic number to satisfy PERF-3. |
| Notes and Issues: | None |

| Use Case ID: | UC-3.1 | | |
|---|---|---|---|
| Use Case Name: | Generate Education Roadmap | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 12 Sep 2026 | Date Last Updated: | 26 Oct 2026 |

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|---|---|
| Actor: | Authenticated Users (Initiating Actor). |
| Description: | Actors are able to generate an education roadmap using both the rule-based method and the AI-generated method, and save a resulting plan in their user account. |
| Preconditions: | - The actor is authenticated.<br>- The actor is on the roadmap page. |
| Postconditions: | - The preferred plan is saved as the roadmap; or<br>- No plan is saved and the system shows an error. |
| Priority: | High |
| Frequency of Use: | Medium |
| Flow of Events: | 1. The authenticated user navigates to the roadmap page.<br>2. The system queries for the saved preferences of the authenticated user.<br>3. The system displays the target transition stage selector with the transition stage preselected using the saved preferences.<br>4. The authenticated user clicks on the “Generate Roadmap” button.<br>5. The system executes the rule set for the selected transition stage and displays the rule-based plan.<br>6. The system invokes the module to create the AI-generated plan asynchronously and displays it.<br>7. The authenticated user clicks on the “Save Roadmap” button of their preferred plan.<br>8. The system stores the preferred plan in the database.<br>9. The system displays a notification confirming the action. |
| Alternative Flows: | - AF-S2: If no postal code is found, the system creates a popup to notify the authenticated user of the missing field with a link to the profile page (UC-2.4). The use case ends.<br>- AF-S3: If the authenticated user changes the target transition stage, the system displays the relevant fields and updates the saved preferences inline. Resume with S4.<br>- AF-S6: If the module errors or exceeds 5s, the system displays “Unable to generate AI recommendation at this time. Please try again.” with a “Retry AI Generation” button in the panel. Resume with S6.<br>- AF-S7: If there is an existing roadmap, the system displays a confirmation dialog to confirm overwrite. Resume with S8. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | - UC-2.4: View User Profile (AF-S3) |
| Special Requirements: | - PERF-6: The system shall display the rule-based plan within 2s of generation.<br>- PERF-7: The system shall display the AI-generated plan within 15s of generation. |

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| | - QUAL-1: The roadmap view shall be responsive and adjust to the screen size of the user.<br>- QUAL-2: The system shall show the inline errors within 0.3s.<br>- QUAL-6: The system shall render the rule-based plan without waiting for the AI-generated plan.<br>- QUAL-7: The system shall display module errors within the panel without reloading the page. |
| Assumptions: | 1. A dedicated rule set exists for every transition stage.<br>2. Each user account holds only one saved roadmap at any time. |
| Notes and Issues: | None |

| Use Case ID: | UC-3.2 | | |
|---|---|---|---|
| Use Case Name: | View Saved Roadmap | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 12 Sep 2026 | Date Last Updated: | 12 Sep 2026 |

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|---|---|
| Actor: | Authenticated Users (Initiating Actor). |
| Description: | Actors are able to view the roadmap saved in their user account. |
| Preconditions: | - The actor is authenticated.<br>- The actor is on the roadmap page. |
| Postconditions: | - The saved roadmap is displayed; or<br>- An empty state is displayed. |
| Priority: | High |
| Frequency of Use: | Medium |
| Flow of Events: | 1. The authenticated user navigates to the roadmap page.<br>2. The system queries for the roadmap of the authenticated user.<br>3. The system displays the roadmap containing the following:<br>  o Target Transition Stage<br>  o School Choices<br>4. The system displays the “Generate Roadmap”, “Edit Roadmap” and “Delete Roadmap” buttons. |
| Alternative Flows: | - AF-S2: If no roadmap is saved, the system displays an empty state with a “Generate Roadmap” button. The use case ends. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | - QUAL-1: The roadmap view shall be responsive and adjust to the screen size of the user. |
| Assumptions: | 1. Each user account holds only one saved roadmap at any time. |
| Notes and Issues: | None |

| Use Case ID: | UC-3.3 | | |
|---|---|---|---|
| Use Case Name: | Edit Saved Roadmap | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 12 Sep 2026 | Date Last Updated: | 12 Sep 2026 |

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|---|---|
| Actor: | Authenticated Users (Initiating Actor). |
| Description: | Actors are able to edit the roadmap saved in their user account. |
| Preconditions: | - The actor is authenticated.<br>- The actor has a saved roadmap.<br>- The actor is on the roadmap page. |
| Postconditions: | - The saved roadmap is updated; or<br>- The roadmap update fails and the system shows a popup error. |
| Priority: | Medium |
| Frequency of Use: | Low |
| Flow of Events: | 1. The authenticated user clicks on the “Edit Roadmap” button of the saved roadmap.<br>2. The system displays a dropdown for each school choice.<br>3. The authenticated user modifies the school choices.<br>4. The authenticated user clicks on the “Save Changes” button.<br>5. The system sanitizes user input.<br>6. The system updates the saved roadmap and displays the updated roadmap. |
| Alternative Flows: | - AF-S3: If the authenticated user clicks on the “Cancel” button, the system discards the changes. The use case ends. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | - SEC-4: The system shall sanitize the user input on the backend. |
| Assumptions: | 1. The system does not verify if the school is a suitable fit.<br>2. The dropdown is populated from the allowed range of values, so invalid references cannot be submitted from the form. |
| Notes and Issues: | None |

| Use Case ID: | UC-3.4 | | |
|---|---|---|---|
| Use Case Name: | Delete Saved Roadmap | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 12 Sep 2026 | Date Last Updated: | 12 Sep 2026 |

| Actor: | Authenticated Users (Initiating Actor). |
|---|---|
| Description: | Actors are able to delete the roadmap saved in their user account. |
| Preconditions: | - The actor is authenticated.<br>- The actor has a saved roadmap.<br>- The actor is on the roadmap page. |
| Postconditions: | - The roadmap is deleted from the user account; or<br>- The roadmap deletion fails and the system shows a popup error. |
| Priority: | Medium |
| Frequency of Use: | Low |
| Flow of Events: | 1. The authenticated user clicks on the “Delete Roadmap” button of the saved roadmap.<br>2. The system displays a confirmation dialog.<br>3. The authenticated user confirms the deletion.<br>4. The system removes the roadmap from the database.<br>5. The system displays the empty state of the roadmap page. |
| Alternative Flows: | - AF-S3: If the authenticated user cancels the confirmation dialog, the system closes the dialog. The use case ends. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | None |
| Assumptions: | 1. There is no undo option after deleting a roadmap. |
| Notes and Issues: | None |

| Use Case ID: | UC-4.1 | | |
|---|---|---|---|
| Use Case Name: | Submit School Review | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 12 Sep 2026 | Date Last Updated: | 12 Sep 2026 |

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|---|---|
| Actor: | Authenticated Users (Initiating Actor). |
| Description: | Actors are able to submit a review on a school profile page. |
| Preconditions: | - The actor is authenticated.<br>- The actor is on a school profile page. |
| Postconditions: | - The review is created and visible from the school profile; or<br>- The review creation fails and the system shows an inline error. |
| Priority: | Medium |
| Frequency of Use: | Low |
| Flow of Events: | 1. The authenticated user navigates to a school profile page.<br>2. The system checks for a review by the authenticated user.<br>3. The system displays a review text area with a word counter.<br>4. The authenticated user types a review into the review text area.<br>5. The authenticated user clicks on the “Submit Review” button.<br>6. The system sanitizes user input.<br>7. The system checks that the review is not empty.<br>8. The system saves the review and refreshes the view. |
| Alternative Flows: | - AF-S2: If a school has already been reviewed, the system displays their existing review with the “Edit Review” and “Delete Review” buttons instead. The use case ends.<br>- AF-S4: If the review reaches 200 words, the text area blocks additional keystrokes. Resume with S5.<br>- AF-S7: If the review is blank or consists entirely of spaces, an inline error "Review text cannot be blank or consist entirely of spaces." will be displayed on the form. Return to S4. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | - BR-6: The system shall allow only one review per school from a single user account.<br>- QUAL-2: The system shall show the inline errors within 0.3s.<br>- SEC-4: The system shall sanitize the user input on the backend. |
| Assumptions: | 1. The reviews are displayed immediately without moderation.<br>2. The feature is hidden from guest users. |
| Notes and Issues: | Considerations<br>1. A moderation flow for user review. |

| Use Case ID: | UC-4.2 | | |
|---|---|---|---|
| Use Case Name: | View School Reviews | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 12 Sep 2026 | Date Last Updated: | 12 Sep 2026 |

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|---|---|
| Actor: | All Users (Initiating Actor). |
| Description: | Actors are able to view all reviews on a school profile page. |
| Preconditions: | - The actor is on a school profile page. |
| Postconditions: | - The reviews are displayed; or<br>- An empty state is displayed. |
| Priority: | Medium |
| Frequency of Use: | High |
| Flow of Events: | 1. The user navigates to a school profile page.<br>2. The system retrieves all the reviews for a school.<br>3. The system displays the following information for each review:<br>o Reviewer Username<br>o Submission Time<br>o Review Content |
| Alternative Flows: | - AF-S2: If there are no reviews of the school, the system displays an empty state "No reviews yet". The use case ends. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | - PERF-1: The system shall load and render the school reviews within 2 seconds on visit or query.<br>- QUAL-1: The review section shall be responsive and adjust to the screen size of the user. |
| Assumptions: | 1. Reviews are sorted from newest to oldest.<br>2. Reviews are shown to guest users. |
| Notes and Issues: | Considerations<br>1. Pagination if there are too many reviews. |

| Use Case ID: | UC-4.3 | | |
|---|---|---|---|
| Use Case Name: | Edit School Review | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 12 Sep 2026 | Date Last Updated: | 12 Sep 2026 |

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|---|---|
| Actor: | Authenticated Users (Initiating Actor). |
| Description: | Actors are able to edit their review of a school profile page. |
| Preconditions: | - The actor is authenticated.<br>- The actor has reviewed the school.<br>- The actor is on a school profile page. |
| Postconditions: | - The review is updated and visible from the school profile; or<br>- The review update fails and the system shows an inline error. |
| Priority: | Low |
| Frequency of Use: | Low |
| Flow of Events: | 1. The authenticated user clicks on the “Edit Review” button of their own review.<br>2. The system displays the review text area populated with the existing review and a word counter.<br>3. The authenticated user modifies the review.<br>4. The authenticated user clicks on the “Update Review” button.<br>5. The system sanitizes user input.<br>6. The system checks that the review is not empty.<br>7. The system updates the review and refreshes the view. |
| Alternative Flows: | - AF-S3: If the review reaches 200 words, the text area blocks additional keystrokes. Resume with S4.<br>- AF-S4: If the authenticated user clicks on the “Cancel” button, the system discards the changes. The use case ends.<br>- AF-S6: If the review is blank or consists entirely of spaces, an inline error "Review text cannot be blank or consist entirely of spaces." will be displayed on the form. Return to S3. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | - QUAL-2: The system shall show the inline errors within 0.3s.<br>- SEC-4: The system shall sanitize the user input on the backend. |
| Assumptions: | 1. The edit button is shown only on the review created by the authenticated user. |
| Notes and Issues: | None |

| Use Case ID: | UC-4.4 | | |
|---|---|---|---|
| Use Case Name: | Delete School Review | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 12 Sep 2026 | Date Last Updated: | 12 Sep 2026 |

| Actor: | Authenticated Users (Initiating Actor). |
|---|---|
| Description: | Actors are able to delete their review of a school profile page. |
| Preconditions: | - The actor is authenticated.<br>- The actor has reviewed the school.<br>- The actor is on a school profile page. |
| Postconditions: | - The review is deleted from the school profile page; or<br>- The review deletion fails and the system shows a popup error. |
| Priority: | Low |
| Frequency of Use: | Low |
| Flow of Events: | 1. The authenticated user clicks on the “Delete Review” button of their own review.<br>2. The system displays a confirmation dialog.<br>3. The authenticated user confirms the deletion.<br>4. The system removes the review from the database.<br>5. The system displays the review text area. |
| Alternative Flows: | - AF-S3: If the authenticated user cancels the confirmation dialog, the system closes the dialog. The use case ends. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | None |
| Assumptions: | 1. The delete button is shown only on the review created by the authenticated user. |
| Notes and Issues: | None |

| Use Case ID: | UC-5.1 | | |
|---|---|---|---|
| Use Case Name: | Update School Database | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 12 Sep 2026 | Date Last Updated: | 26 Oct 2026 |

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|---|---|
| Actor: | Administrators (Initiating Actor). |
| Description: | Actors are able to manually update the school database by uploading a CSV file containing the latest school dataset obtained from external sources such as “data.gov.sg”. |
| Preconditions: | - The actor is authenticated as an administrator.<br>- The actor is on the admin panel. |
| Postconditions: | - The school database is updated with the latest dataset; or<br>- The school database is unchanged and the system shows a popup error. |
| Priority: | High |
| Frequency of Use: | Low |
| Flow of Events: | 1. The administrator navigates to the admin panel.<br>2. The system displays a file upload control and a disabled “Update Database” button.<br>3. The administrator selects the latest school dataset as a CSV file.<br>4. The system validates the file, displays its name, and enables the “Update Database” button.<br>5. The administrator clicks on the “Update Database” button.<br>6. The system disables further interactions with the file upload control and “Update Database” button.<br>7. The system parses and standardizes the CSV records.<br>8. The system creates a transaction to bulk write the records.<br>9. The system commits the transaction, re-enables the control and button, and displays “Database updated successfully with the latest Singapore school data”. |
| Alternative Flows: | - AF-S3: If no CSV file is selected, the system displays “Please select a CSV file before updating the database.” Return to S3.<br>- AF-S4: If selected file is invalid, empty, or exceeds the size limit, the system displays “Invalid file. Please upload a non-empty .csv file within the size limit.” Return to S3.<br>- AF-S7: If an error occurs during parsing or standardization, the system aborts the update, rolls back any pending changes, re-enables the controls, and displays “Database Update Failed: Unable to process dataset format. Please check source data and try again.” Return to S3. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | - BR-5: The system shall allow database update functions only to administrator accounts.<br>- PERF-8: The system shall complete the operation within 10s. |
| Assumptions: | 1. The administrator obtains a CSV file that follows the expected dataset schema. |

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|---|---|
| | 2. The administrator role is assigned directly in the database. |
| Notes and Issues: | None |

| Use Case ID: | UC-5.2 | | |
|---|---|---|---|
| Use Case Name: | Export School Database | | |
| Created By: | Max | Last Updated By: | Max |
| Date Created: | 26 Oct 2026 | Date Last Updated: | 26 Oct 2026 |

| Actor: | Administrators (Initiating Actor). |
|---|---|
| Description: | Actor are able to export the entire school database, including all school records and associated details into a CSV file for backup. |
| Preconditions: | - The actor is authenticated as an administrator.<br>- The actor is on the admin panel. |
| Postconditions: | - A CSV file containing the current school database; or<br>- No file is downloaded and the system shows a popup error. |
| Priority: | Medium |
| Frequency of Use: | Low |
| Flow of Events: | 1. The administrator navigates to the admin panel.<br>2. The administrator clicks on the “Export Database” button.<br>3. The system queries the database to retrieve all school records and associated details.<br>4. The system formats the retrieved records into a CSV file named “pathsg_schools_YYYYMMDD.csv” using the current date.<br>5. The system initiates the download of the CSV file to the administrator’s local machine. |
| Alternative Flows: | - AF-S3: If the school database is empty, the system aborts the export and displays “Export Failed: Unable to retrieve school data at this time.” The use case ends. |
| Exceptions: | - EX-1: The database is unreachable. The system will create a popup to notify the user of the error. |
| Includes: | None |
| Special Requirements: | - PERF-9: The system shall complete the export operation within 5s.<br>- SEC-6: The system shall restrict database export function to authenticated administrator accounts. |
| Assumptions: | 1. The administrator obtains a CSV file that follows the expected dataset schema.<br>2. The administrator role is assigned directly in the database. |
| Notes and Issues: | None |